

Practical

The practical that relates to the case study or visit should give candidates the opportunity to be assessed in four skill areas:

- summarising details of a visit or case study
- planning a practical
- implementation and recording of measurements
- analysis of results and drawing conclusions.

The planning, implementation and analysis aspects of the practical work must be carried out individually and under supervision.

The practical should lead to a graph relating two measured variables. Ideally the candidate should then attempt to derive the equation relating the two variables or a relevant quantity to the topic, for example the value of resistivity for a particular material.

Use of ICT

Candidates may use a word processor to produce their summary of the visit or the case study, although they will not gain any extra marks for doing so.

In order to ensure that candidates demonstrate their understanding of the principles and techniques involved in analysing data, the use of ICT, eg spreadsheets, may not be used for analysing data for this unit.

Draft work

Students should carry out a variety of practical work during the course so that they develop the necessary skills to succeed in this unit. They should **not**, therefore, submit draft assessment work for checking. However, teachers should check students' plans for health and safety issues.

Work submitted for this unit must **not** be returned to students for them to improve it.

How Science Works

This unit will cover the following aspects of *how science works* as listed in *Appendix 6: How science works* 2, 3, 4, 5, 6, 8, and 9.

9.2 Assessment information

Summary of visit or case study	Students should produce a brief summary of the case study or physics-based visit as homework. It is recommended that students word process this part of the assessment. The summary should be between 500–600 words.
Plan	Students may be given the title of the experiment that they are to plan and carry out in advance. The plan should be produced under supervised conditions in class in the students' own handwriting. Students should not take any documents into the classroom as they should have gained sufficient experience of planning practical work during normal practical lessons. Teachers should collect in the plan at the end of the session to check for health and safety issues. The plan will need to be returned to students so that they can carry out their plan. At this stage teachers could either: i) photocopy the plan, mark the plan if it is to be internally assessed, and provide students with the photocopy in the laboratory so that they can carry out their plan ii) collect in the plan, not mark it and return it to students in the laboratory under supervised conditions so that they can carry out their plan.
Practical	The practical work should be carried out under supervised conditions in a separate session from the planning session. Unmarked plans should be returned to students so that they can carry out the experiment that they have planned. Students should work individually. If necessary, teachers may allow students to analyse results under supervision in their next lesson. In this situation, teachers must collect in the written work produced by their students. Teachers should not mark the plan or practical work. In the next lesson, the documents should be returned to students under supervised conditions for analysis. Students should not have access to any other sources of information while they are completing the analysis of their results